Global Health and Demography

Week 2: Contemporary Mortality and Fertility Transition

Recap: Demographic changes and Global Health

- **DTM은 실제로 관찰되는가?**

  - Descriptive validity

- **사망과 출산 감소의 관계는 어떻게 설명되는가?**

  - Explanatory mechanism

- **어떻게 사망과 출산을 줄일 수 있을까?**

  - Mortality Transition (Econ vs Health)
  - Fertility Transition (Econ vs Health)

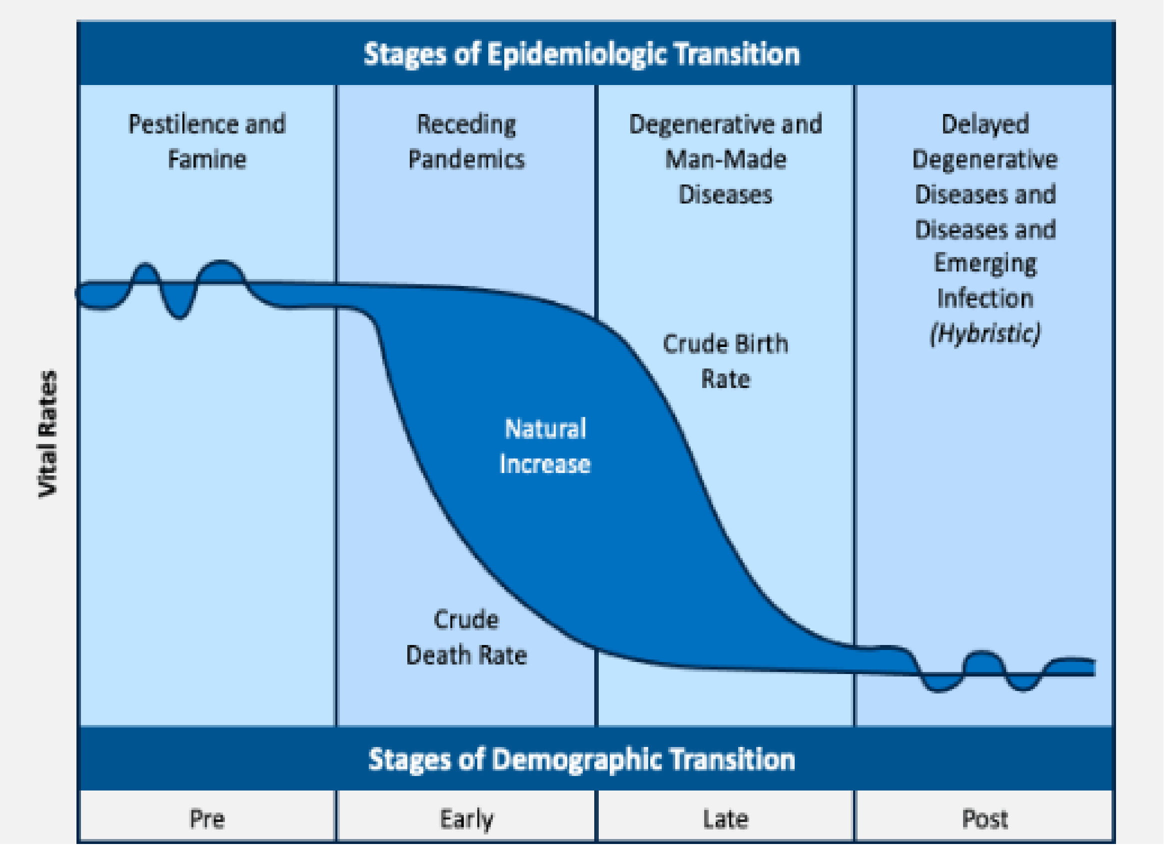
- **인구구조와 인구증가관계**

Recap: Demographic changes and Global Health

Today

- Epidemiologic Transition Theory (ETM)
- Evolution of Measuring Mortality
- Evolution of Measuring Fertility
- Contemporary Mortality & Fertility in Globe

Epidemiological Transition



- **역학변천이론 (Abdel Omran, 1971)**
  - 인구변천모형(DTM)을 바탕으로, 출생·사망률 변화와 연동된 **질병 및 사망 원인의 변화**를 설명하는 이론
- **보건의료 발전에 따른 질병 구조의 변화**
  - 감염성 질환 중심 (Pestilence & Famine) → 만성 비감염성 질환 중심 (Degenerative Diseases)
- **인구학적 의의**
  - 현대 사회의 인구 고령화 및 기대수명 증가 현상 설명

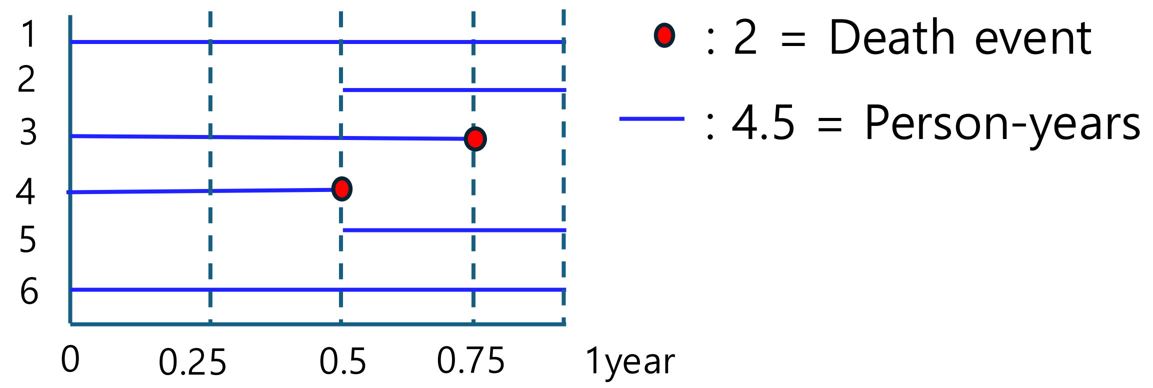
Proportion, Ratio, Rate

1. **Proportion:** 전체 인구 대비 하위 집단의 크기. 분자가 분모에 포함 (서울대 전체 학생중 남학생비율)

2. **Ratio:** 한 집단과 다른 집단의 상대적 크기 비교. 분자와 분모가 독립적 (서울대 남학생 : 서울대 여학생)

3. **Rate:** 시간적 노출(Time exposure)을 고려한 사건의 발생 빈도.

$$\frac{\text{발생한 사건 수 (Events)}}{\text{인구 (Population)} \times \text{시간적 노출량 (Time Exposure)}}$$



Evolution of Measuring Mortality

1. Crude death rate
2. Age specific mortality rates
3. Age standardized morality rates
4. Life expectancy
5. HALE
6. DALY and Burden of Disease

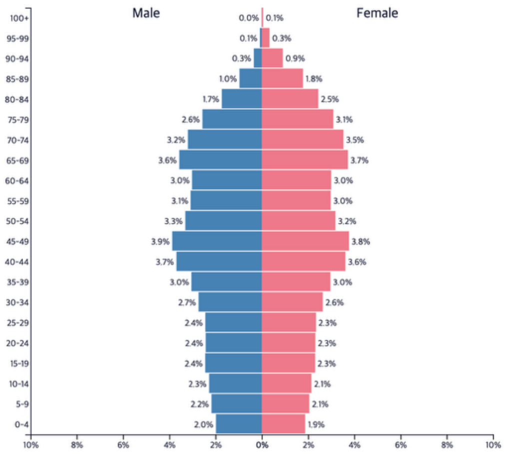
사망

생존

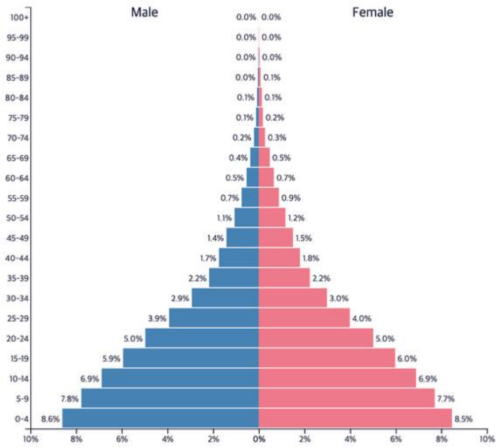
건강손실

Crude Death Rate

Which country is healthier (2018)?



Japan

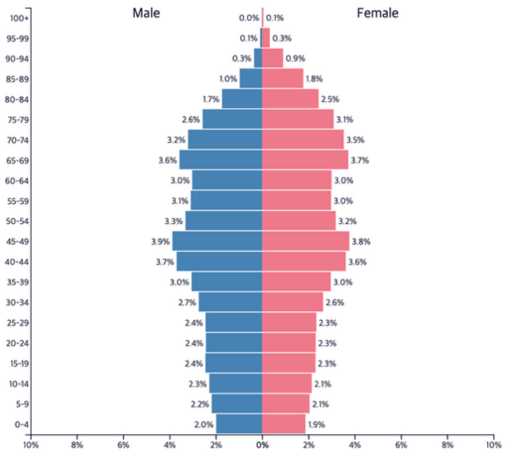


Uganda

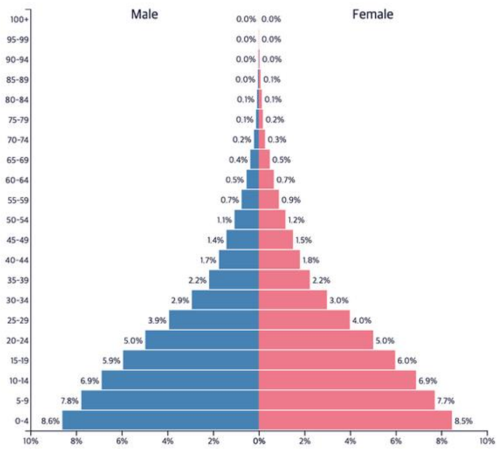
Country	Life Expectancy	CDR
Japan	84 years	**11.30**
Uganda	64 years	**5.62**

Crude Death Rate & Age specific mortality rate

Which country is healthier (2018)?



Japan



Uganda

Country	Life Expectancy	CDR
Japan	84 years	11.30
Uganda	64 years	5.62

Japan

Age	No. of deaths	Population (1,000's)	Age-specific MR
<20	4,403 (**0.3%**)	21,768 (**17.2%**)	0.20
20-59	83,139 (**5.8%**)	61,327 (**48.3%**)	1.36
60+	1,347,942 (**94.9%**)	43,728 (**34.5%**)	20.82
All ages	**1,435,485**	**126,823**	**(CDR) 11.30**

Uganda

Age	No. of deaths	Population (1,000's)	Age-specific MR
<20	92.039 (**38.8%**)	24,501 (**58.0%**)	3.76
20-59	84,702 (**35.7%**)	16,398 (**38.9%**)	5.17
60+	60.591 (**25.5%**)	21,362 (**3.1%**)	44.49
All ages	**237,332**	**42,262**	**(CDR) 5.62**

CDR 한계점

- **국가 간 비교가 어려움**

  1) 인구 연령구조의 영향을 받음

  노인 인구가 많은 국가는 실제 건강수준과 무관하게 CDR이 높게 나타날 수 있음

  2) 어느. 연령에서 사망이 발생하는지 알 수 없음

  영아사망과 노인사망을 구분하지 못함.

  3) 국가 간 직접 비교에 한계

  동일한 CDR이라도 사망 패턴은 크게 다를 수 있음

- **해결방법**

  1) **Age-Specific Mortality Rate :** 연령별 사망율

  2) **Age-Standardized Mortality Rate :** 연령구조 차이를 제거한 후 비교

CDR vs Age Standardized mortality rate

Age	Population (Uganda)	Population (Japan)	Standard Population	Age-specific MR (Uganda)	Age-specific MR (Japan)	Expected dealth (Uganda)	Expected death (Japan)
<20	24,501	21,768	23,135	3.76	0.20	86,988	4,627
20-59	16,398	61,327	38,863	5.17	1.36	200,921	52,853
60+	1,363	43,728	22,546	44.49	30.82	1,003,269	694,470
All ages	**42,262**	**126,823**	**84,544**	**CDR = 5.62**	**CDR = 11.30**	**1,291,178**	**751,950**
						(ASMR) 15.27	**(ASMR) 8.89**

- **CDR suggests Japan is less healthy than Uganda**

  Japan CDR = 11.30 VS. Uganda CDR = 5.62

- **After age standardization, the conclusion is reversed**

  Japan ASMR = 8.89. VS. Uganda ASMR = 15.27

CDR은 전체 인구 대비 사망 수준을 비교하지만, ASMR은 연령구조를 제거한 사망위험 비교 (둘다 필요함)

Death numbers and Risks

	All-age deaths, thousands			Age-standardised death rate per 100 000 population		
	2000	2023	Percentage change, 2000–23	2000	2023	Percentage change, 2000–23
All causes	50 746·1 (50 430·8 to 51 060·8)	60 043·1 (59 045·4 to 61 239·8)	18·3% (16·1 to 20·8)*	1009·0 (1002·5 to 1015·2)	701·5 (690·2 to 714·9)	–30·5% (–31·7 to –29·0)*
Communicable, maternal, neonatal, and nutritional diseases	15 054·6 (14 495·6 to 15 946·2)	10 326·0 (9761·6 to 11 100·0)	–31·1% (–34·7 to –27·2)*	263·4 (252·2 to 280·2)	136·9 (129·9 to 146·0)	–47·8% (–50·4 to –45·2)*
Non-communicable diseases	31 130·5 (30 469·8 to 31 722·1)	44 842·6 (43 824·4 to 46 047·5)	43·9% (39·5 to 48·6)*	666·6 (653·0 to 678·2)	506·2 (494·7 to 519·3)	–24·1% (–26·4 to –21·7)*
Neoplasms	7072·9 (6679·9 to 7356·8)	10 567·9 (9726·9 to 11 166·9)	49·3% (41·2 to 57·8)*	143·9 (135·8 to 150·0)	117·0 (107·7 to 123·5)	–18·8% (–22·6 to –14·4)*

1. Communicable diseases are declining

- 사망자 수: **15.1 → 10.3 million (-31.1%)**
- 연령표준화 사망률: **263.4 → 136.9 (-47.8%)**

2. Non-communicable diseases are increasing

- 사망자 수: **31.1 → 44.8 million (+43.9%)**
- 연령표준화 사망률: **666.6 → 506.2 (-24.1%)**

- NCD의 연령표준화 사망률(ASMR)은 감소(-24.1%)하여 실제 사망위험은 개선되고 있음.
- 그러나 인구고령화와 인구 증가로 인해 NCD 사망자 수는 오히려 증가(+43.9%)하고 있음.
- 의료기술과 보건의료체계의 발전으로 개인의 사망위험은 감소하고 있지만, 고령인구의 증가가 전체 사망자 수 증가시키고 있음 (향후 NCD 사망인구수 감소는 어려움).

Naghavi, Mohsen, et al. "Global burden of 292 causes of death in 204 countries and territories and 660 subnational locations, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023." *The Lancet* 406.10513 (2025): 1811-1872.

Life expectancy (기대수명)

- 사망지표가 '죽음(Death)'에 초점을 둔다면, 기대수명은 '생존(Survival)'에 초점을 둠.

- 기대수명은 '생명표'로 계산함

- 생명표 주요 지표

  l_x: 생존자 수 (survivors)

  d_x: 사망자 수 (deaths)

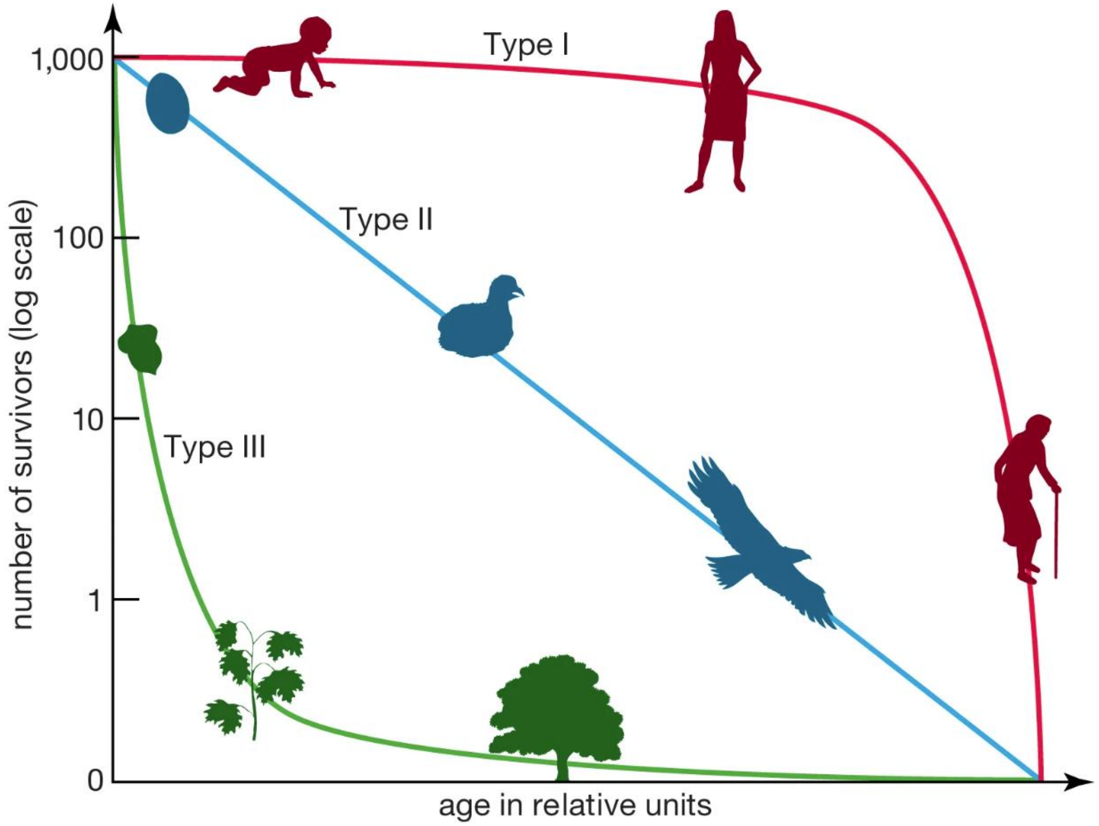
  q_x: 사망확률 (probability of dying)

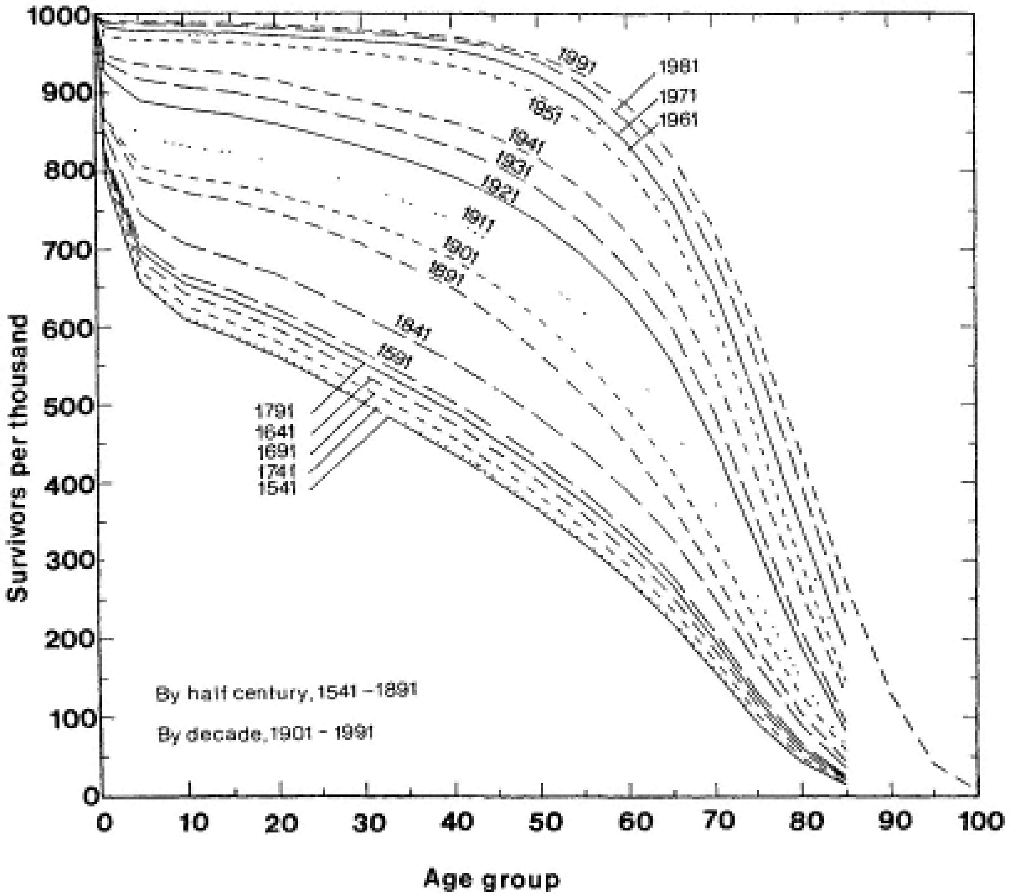
  e_x: 기대수명 (life expectancy)

Age	Interval	nMx	nAx	nqx	lx	ndx	nLx	Sx	Tx	ex
0	1	0.0025	0.1442	0.0025	100000.0000	252.5799	99783.8500	0.9972	8322930.7400	83.2293
1	4	0.0002	1.6454	0.0007	99747.4200	73.7491	398816.0300	0.9993	8223146.8900	82.4397
5	5	0.0001	2.5000	0.0005	99673.6710	45.3860	498254.8900	0.9996	7824330.8600	78.4995
10	5	0.0001	2.5000	0.0003	99628.2850	33.1209	498058.6200	0.9995	7326075.9700	73.5341
15	5	0.0001	2.5000	0.0006	99595.1640	60.3165	497825.0300	0.9993	6828017.3400	68.5577
20	5	0.0002	2.5000	0.0008	99534.8480	84.0615	497464.0800	0.9991	6330192.3100	63.5978
25	5	0.0002	2.5000	0.0010	99450.7860	100.3151	497003.1400	0.9988	5832728.2300	58.6494
30	5	0.0003	2.5000	0.0014	99350.4710	141.2752	496399.1700	0.9981	5335725.0900	53.7061
35	5	0.0005	2.5000	0.0023	99209.1960	227.1024	495478.2200	0.9969	4839325.9200	48.7790
40	5	0.0008	2.5000	0.0038	98982.0930	378.7811	493963.5100	0.9949	4343847.7000	43.8852
45	5	0.0013	2.5000	0.0064	98603.3120	633.1974	491433.5700	0.9916	3849884.1800	39.0442
50	5	0.0021	2.5000	0.0104	97970.1150	1015.6060	487311.5600	0.9859	3358450.6100	34.2804
55	5	0.0036	2.5000	0.0179	96954.5090	1735.2760	480434.3500	0.9756	2871139.0500	29.6133
60	5	0.0063	2.5000	0.0310	95219.2330	2951.0611	468718.5100	0.9561	2390704.7000	25.1074
65	5	0.0118	2.5000	0.0572	92268.1720	5278.7420	448144.0000	0.9303	1921986.1900	20.8304
70	5	0.0173	2.5000	0.0830	86989.4300	7217.6503	416903.0200	0.8868	1473842.1800	16.9428
75	5	0.0315	2.5000	0.1462	79771.7790	11660.3061	369708.1300	0.8121	1056939.1600	13.2495
80	5	0.0537	2.5000	0.2368	68111.4730	16131.0022	300229.8600	0.7049	687231.0300	10.0898
85	5	0.0913	2.5000	0.3715	51980.4710	19311.7572	211622.9600	0.5601	387001.1700	7.4451
90	5	0.1512	2.5000	0.5487	32668.7140	17923.9814	118533.6200	0.3891	175378.2100	5.3684
95	5	0.2393	2.5000	0.7487	14744.7330	11039.7538	46124.2800	0.1886	56844.5900	3.8552
100	NA	0.3742	2.8935	1.0000	3704.9790	3704.9788	10720.3100	0.0000	10720.3100	2.8935

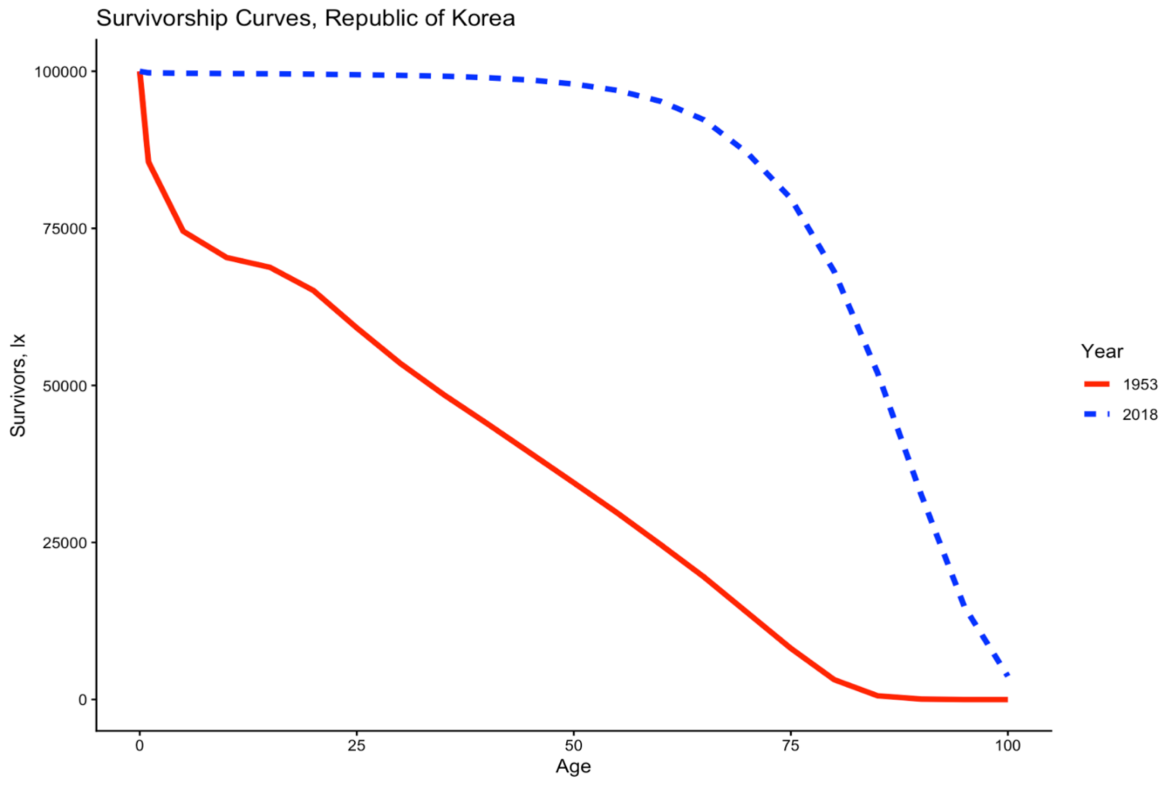
Survival curve (생존곡선)

- 출생한 집단(cohort)이 나이가 증가함에 따라 얼마나 생존하는지를 보여주는 곡선
- 사망을 직접 측정하는 것이 아니라 생존 경험(survival experience)을 시각화한 지표





한국 생존곡선 (1953 vs 2018)



- 1953년에는 영아·아동 사망이 매우 높고, 청-중장년에도 사망이 높았음 (6.25 전쟁)
- 2018년에는 대부분의 출생자가 노년기까지 생존하며, 사망이 고령층에 됨

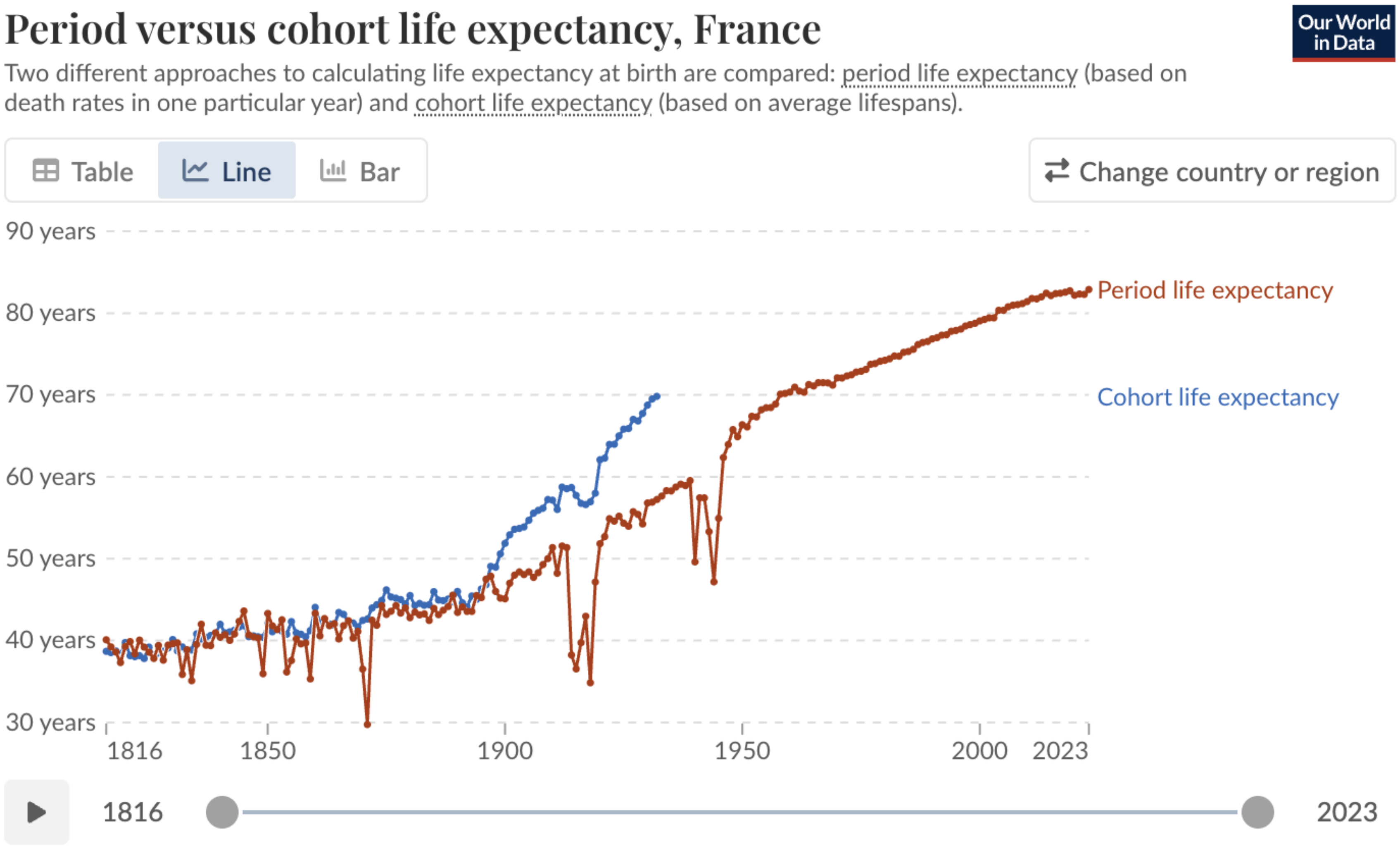
LE 한계점

- **Definition:** The average number of years that a person could expect to live, if he or she were to pass through the remaining years of life exposed to age-specific death rates prevailing at the time, for a specific year.

1) Period vs Cohort

2) 건강하게 생존하는지, 아프게 생존하는지 알 수 없음.

Period vs Cohort LE



Healthy LE (HALE)

- 건강수명(HALE)은 기대수명에서 질병·장애로 인해 건강하지 못한 기간을 제외한 건강한 생존연수

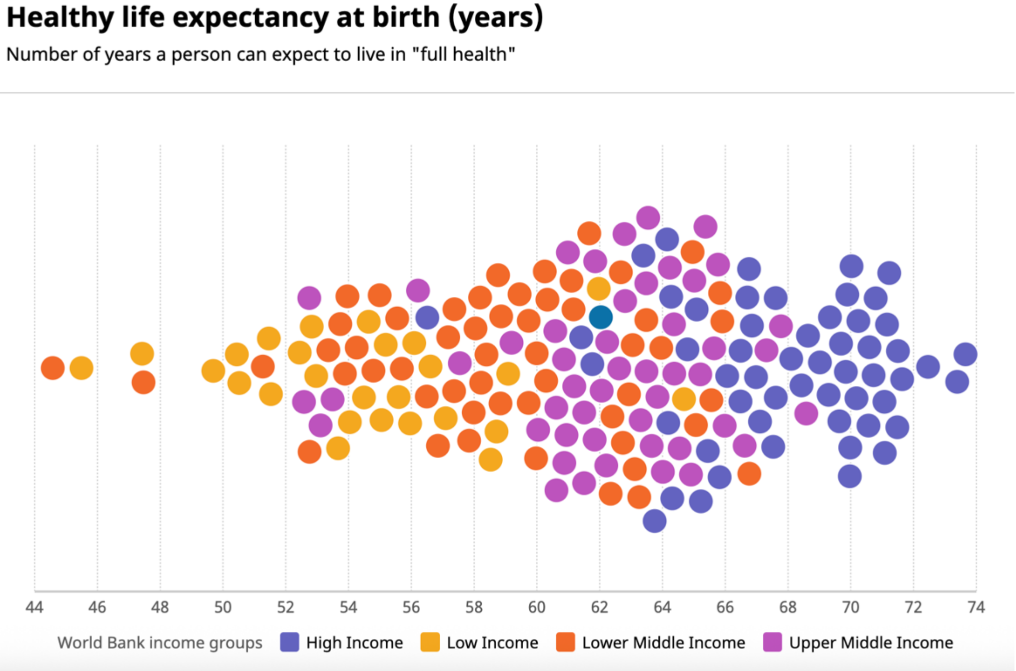
동아일보

오피니언 정치 경제 국제 사회 문화 연예 스포츠 헬스동아

경제

작년 출생아 기대수명 83.7년 '역대 최고'… 건강수명 66.4년 '유병장수'

뉴시스(신문) | 입력 2025-12-03 12:40



https://data.who.int/dashboards/global-progress/hale

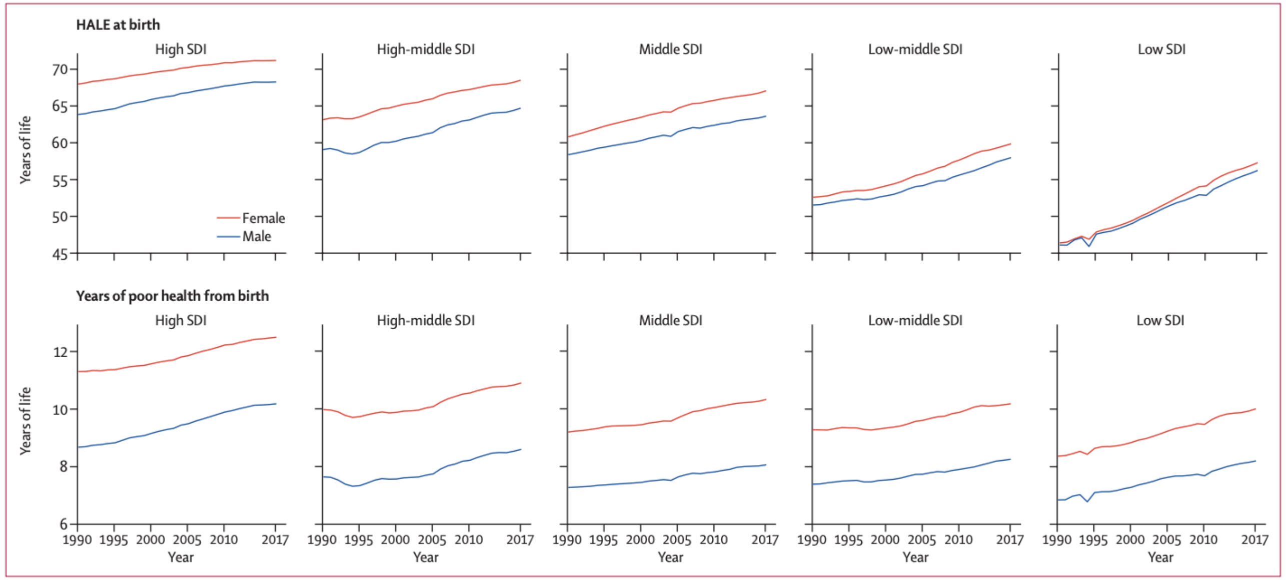
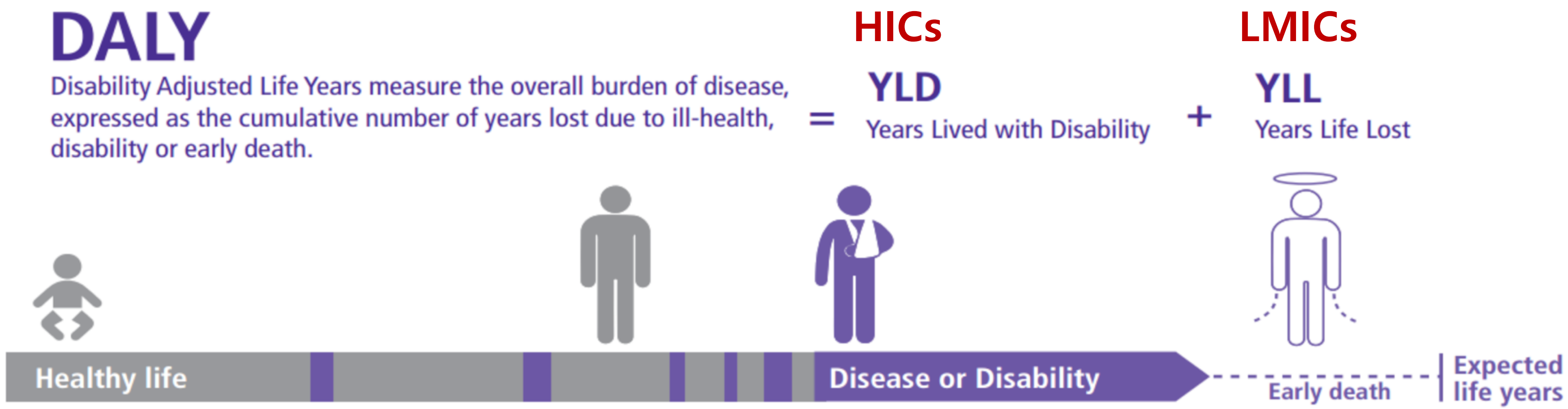


Figure 1: **Trends of HALE at birth and years in poor health from birth by SDI quintile and sex, 1990–2017**
HALE=healthy life expectancy. SDI=Socio-demographic Index.

Kyu, Hmwe Hmwe, et al. "Global, regional, and national disability-adjusted life-years (DALYs) for 359 diseases and injuries and healthy life expectancy (HALE) for 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017." *The Lancet* 392.10159 (2018): 1859-1922.

DALY (Disability Adjusted Life Years)

- 1990년대 초까지 보건의 목표는 '사망을 줄이는 것' 이었음.
- 1990년대 후반 부터 보건의 목표는 '건강하게 오래 사는 것' 으로 중점을 두고, 사망뿐 아니라 건강손실까지 측정하기 시작함 (만성질환, 난청, 시력손실등 장애 측정)



Source : Wiki Commons

DALYs = Years of life lost due to premature mortality (YLL)
+ Years lived with disability (YLD)

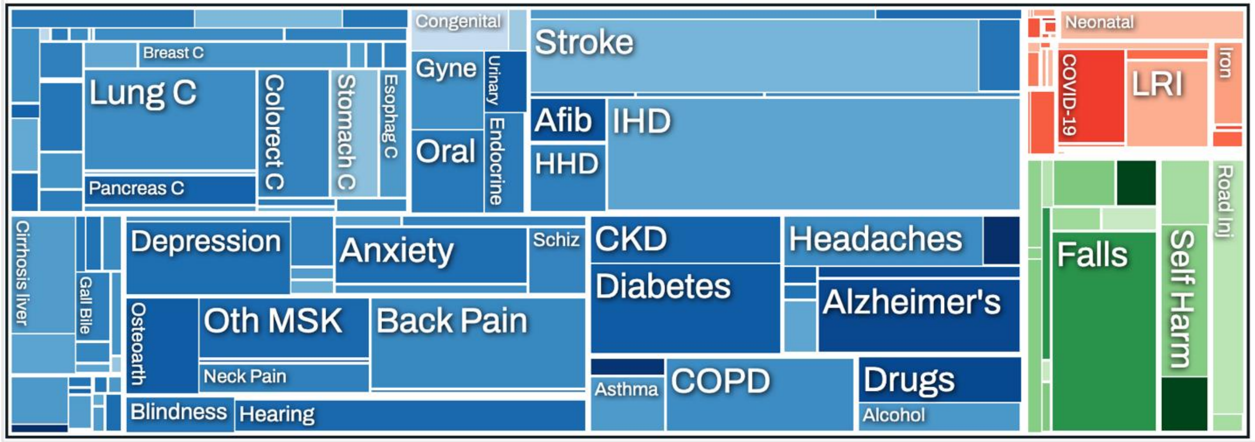
Year lived with Disabilities (YLD)

	Cause	YLDs 2021			Percentage change, 2010–21	
		Percentage of all-cause YLDs	Count (millions)	Age-standardised rate (per 100 000)	YLD count	Age-standardised rate
1	Low back pain	7·7% (6·5 to 9·1)	70·2 (50·2 to 94·1)	832·2 (595·9 to 1115·3)	19·4% (17·9 to 20·8)	–2·4% (–2·9 to –1·8)
2	Depressive disorders	6·2% (5·0 to 7·8)	56·3 (39·3 to 76·5)	681·2 (475·3 to 924·0)	36·5% (31·7 to 41·7)	16·4% (11·9 to 21·3)
3	Headache disorders	5·2% (1·2 to 10·4)	48·0 (9·80 to 101)	588·4 (117·6 to 1245·7)	14·4% (12·7 to 17·8)	0·3% (–1·1 to 1·4)
4	Age-related and other hearing loss	4·9% (3·9 to 5·9)	44·4 (30·7 to 62·0)	525·9 (364·2 to 731·9)	31·6% (30·0 to 33·1)	2·5% (1·7 to 3·3)
5	Other musculoskeletal disorders	4·8% (3·6 to 6·2)	43·0 (29·6 to 59·2)	507·2 (349·5 to 698·0)	28·7% (26·7 to 30·8)	6·3% (5·6 to 7·0)
6	Anxiety disorders	4·7% (3·6 to 6·0)	42·5 (29·4 to 57·7)	524·3 (363·3 to 716·2)	33·7% (30·7 to 37·0)	16·7% (14·0 to 19·8)
7	Diabetes mellitus	4·5% (3·9 to 5·2)	41·2 (29·0 to 56·5)	477·0 (336·1 to 653·3)	62·9% (60·3 to 65·8)	25·9% (24·0 to 28·1)
8	Dietary iron deficiency	3·6% (2·8 to 4·2)	32·3 (21·8 to 46·5)	423·7 (285·3 to 611·0)	3·2% (0·9 to 5·2)	–7·3% (–9·4 to –5·5)
9	Blindness and vision loss	3·2% (2·4 to 4·4)	29·2 (19·0 to 42·9)	342·8 (224·2 to 503·6)	32·9% (27·4 to 38·6)	2·6% (–2·5 to 7·7)
10	Gynaecological diseases	3·0% (2·5 to 3·7)	27·4 (19·1 to 38·2)	334·0 (232·7 to 467·5)	15·3% (14·0 to 16·8)	1·4% (0·6 to 2·3)

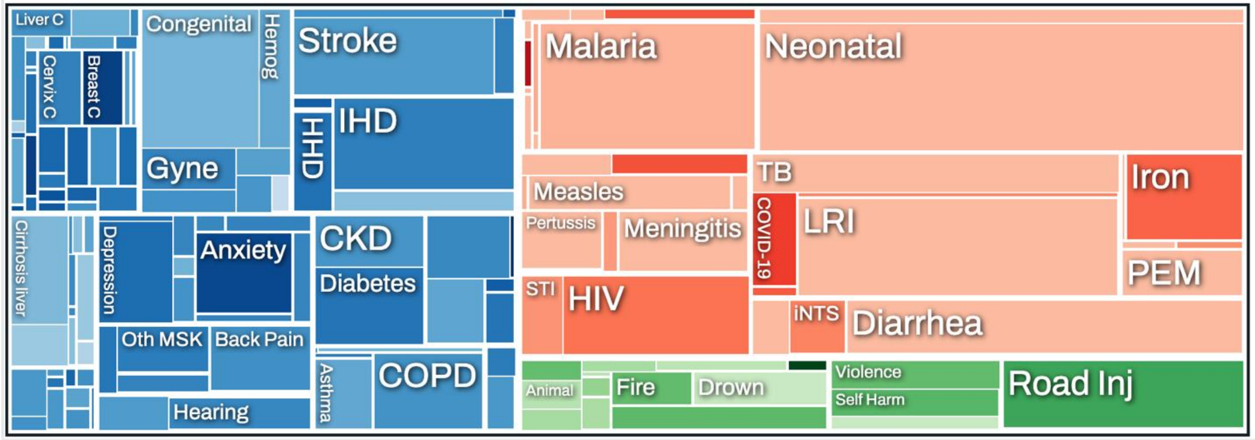
Ferrari, Alize J., et al. "Global incidence, prevalence, years lived with disability (YLDs), disability-adjusted life-years (DALYs), and healthy life expectancy (HALE) for 371 diseases and injuries in 204 countries and territories and 811 subnational locations, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021." *The Lancet* 403.10440 (2024): 2133-2161

Global Burden of disease (GBD)

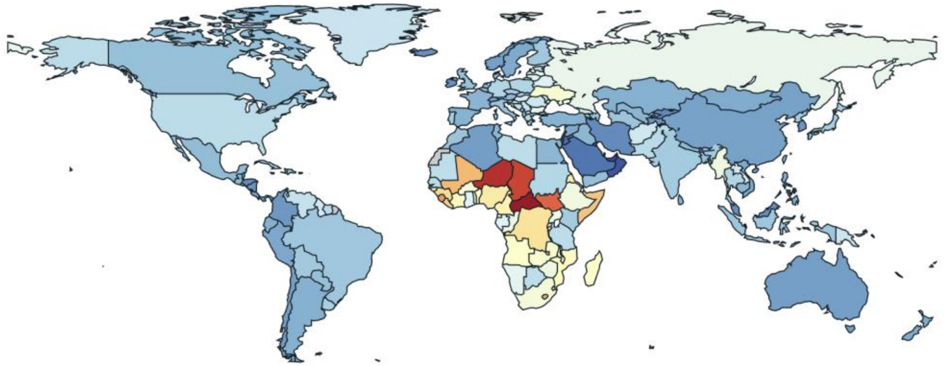
High SDI
Both sexes, All ages, 2023, DALYs



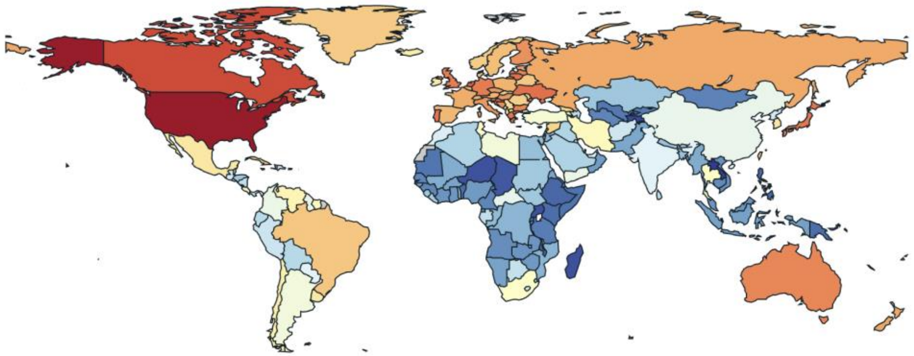
Low SDI
Both sexes, All ages, 2023, DALYs



All causes
Both sexes, All ages, 2023, DALYs per 100,000

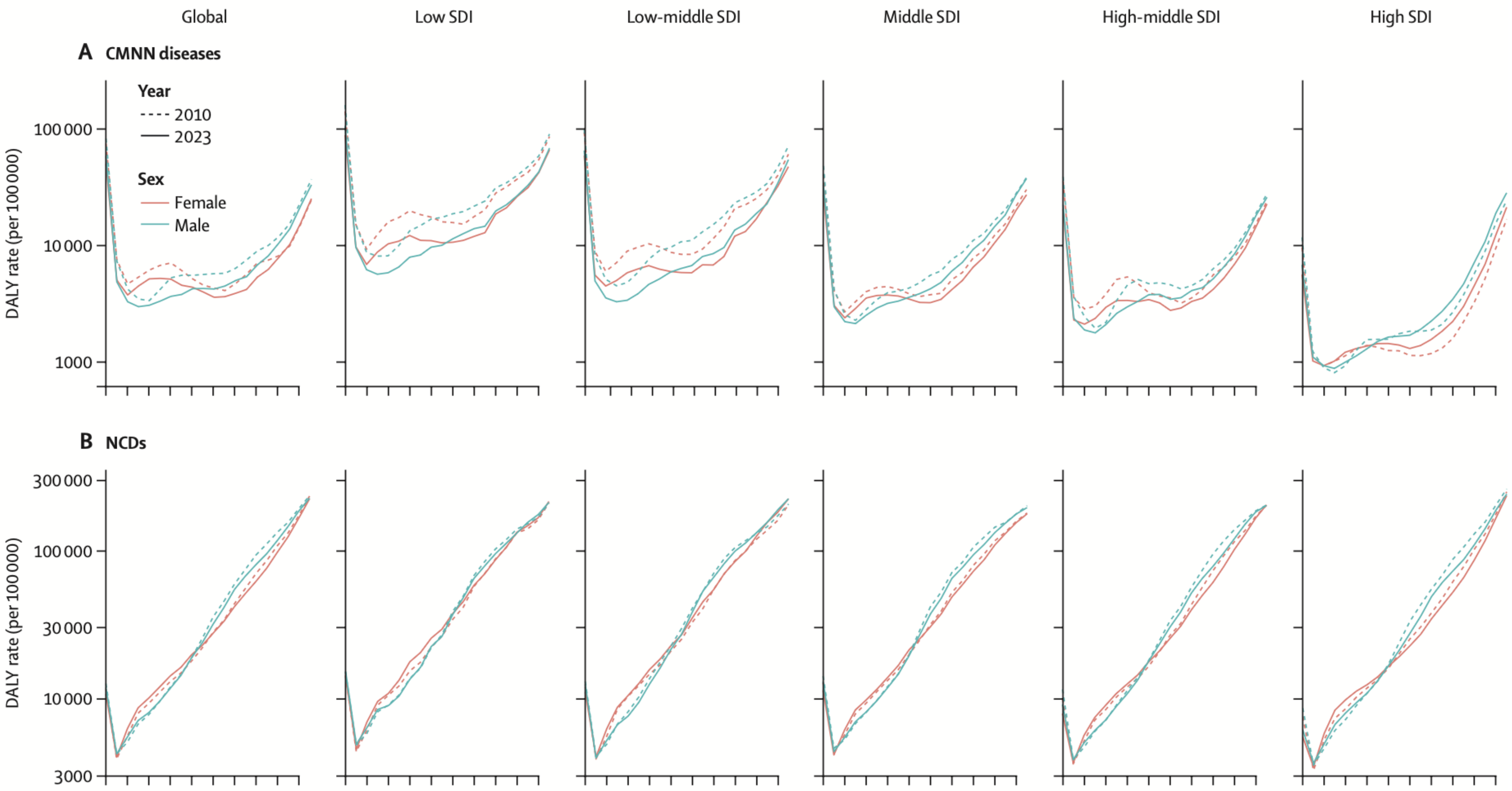


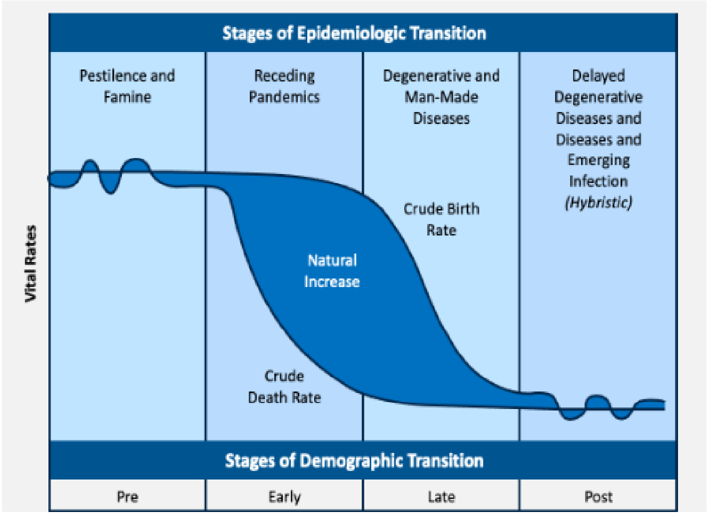
All causes
Both sexes, All ages, 2023, YLDs per 100,000



https://vizhub.healthdata.org/gbd-compare/

(GBD) Age specific DALY





Evolution of Measuring Fertility

1. Crude Birth Rate
2. General Fertility Rate
3. Age-specific Fertility Rates
4. The Total Fertility Rate
5. Determinant of fertility

Crude Birth Rate (조출생률)

- 출생아 수를 전체 인구로 나눈 지표

$$\text{Crude Birth Rate} = \frac{\text{Total births in a stated period}}{\begin{array}{c}\text{Total population over that period} \\ \text{(Mid-year Population)}\end{array}}$$

- **한계점**

1) 전체 인구를 분모로 사용 해서 실제 출생의 주체인 **가임기 여성(15–49세)** 을 고려하지 못함

2) 동일한 출산행태라도 인구구조에 따라 CBR이 달라질 수 있음

General fertility rate

- 출생아 수를 가임기 여성(15–49세) 수로 나눈 지표

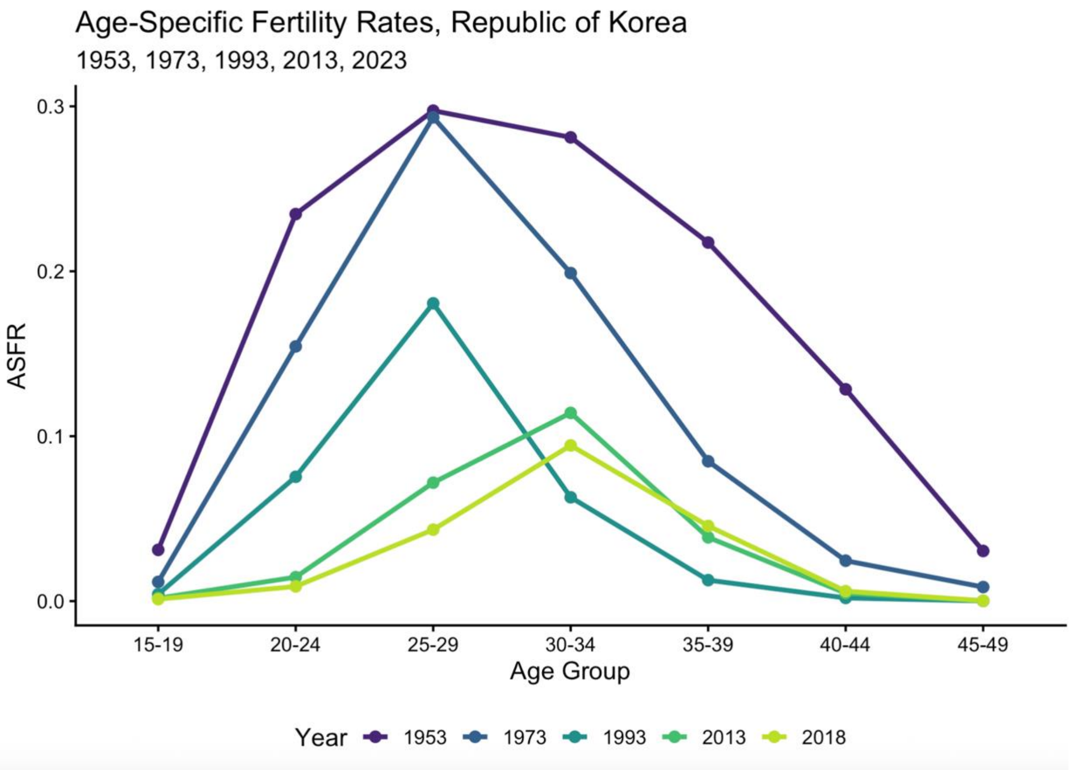
$$\text{General Fertility Rate} = \frac{\text{births in a stated period}}{\text{number of women aged 15 - 49 in the same period}}$$

- **한계점**

1) 가임기 여성 내부의 연령구조를 고려하지 못함

2) 출산은 특정 연령대(주로 20~39세)에 집중됨

Age specific fertility rate



$$\text{ASFR at age} = \frac{\text{births to women at specific ages}}{\text{number of women at specific ages}}$$

- 장점
  1) 연령구조의 고려하여 비교 가능
  2) 출산 시기(Tempo)와 출산 수준(Quantum)을 구분
  3) TFR 계산의 기본 자료

year	15-19	20-24	25-29	30-34	35-39	40-44	45-49
1953	31.1	234.7	297.4	281.2	217.4	128.4	30.4
1973	11.6	154.4	293.3	198.9	84.8	24.5	8.5
1993	4.2	75.4	180.6	62.9	12.7	1.9	0.1
2013	1.8	14.6	71.8	114.1	38.7	4.8	0.2
2018	1.2	8.9	43.3	94.4	45.6	6	0.2

Total fertility rate

- Definition: The number of children that would be born to a woman if she were to live to the end of her childbearing years and bear children in accordance with age-specific fertility rates currently observed.

https://databank.worldbank.org/metadataglossary/world-development-indicators/series/SP.DYN.TFRT.Q1

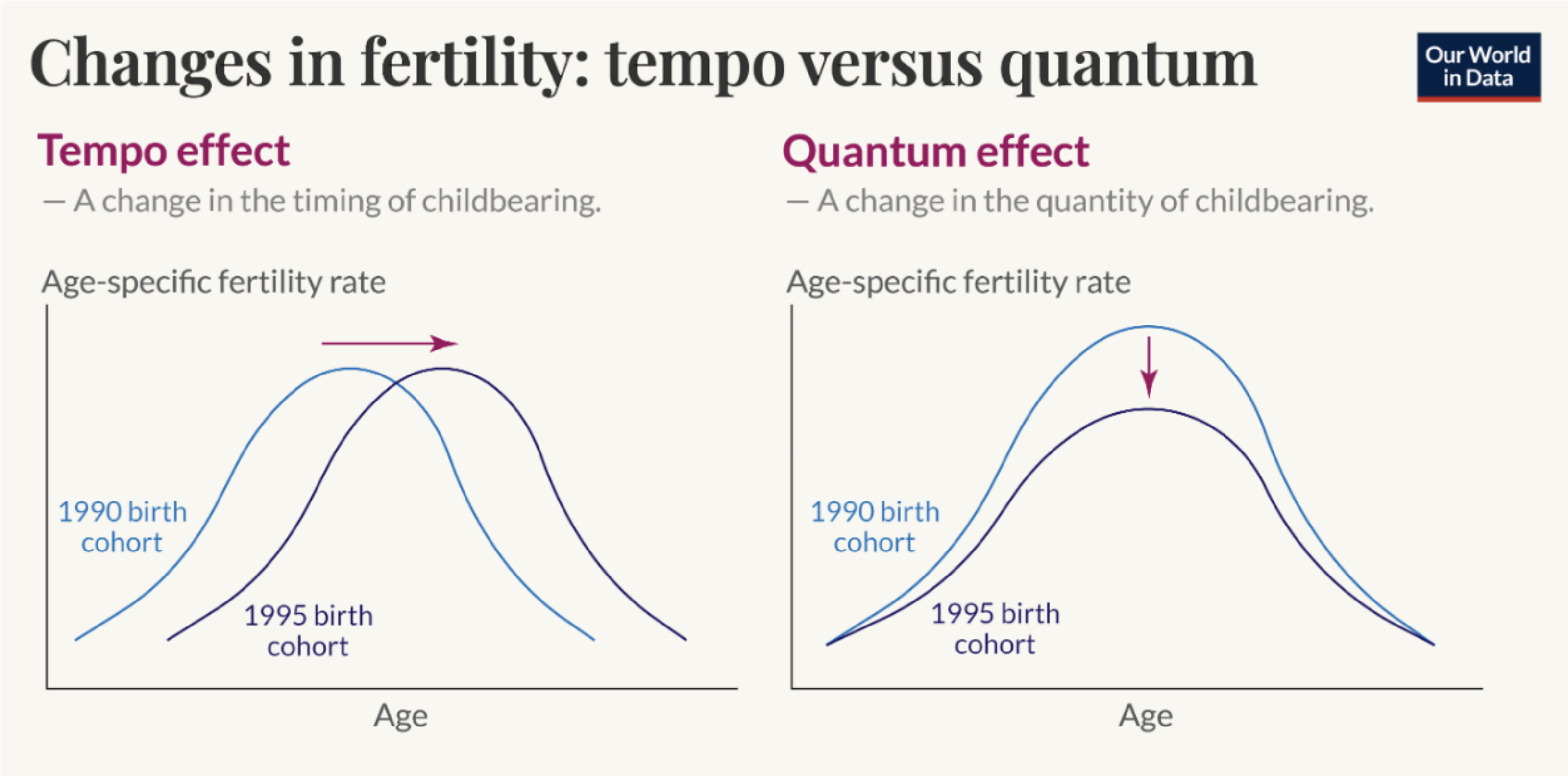
How is TFR calculated?

$$TFR = 5 \times \sum ASFR_x$$

5 = width of each age group (15–19, 20–24, …, 45–49)

- **출산의 주체인 여성의 출산행태를 반영:** 연령별 출산율(ASFR)을 기반으로 계산
- **연령구조의 영향을 최소화:** CBR, GFR보다 국가 간 비교에 적합
- **국가 및 시계열 비교 가능 :**인구학에서 가장 널리 사용되는 출산 지표
- **한계점:** Period indicator

Total fertility rate – Tempo & Quantum effects



- **Tempo Effect (출산 시기의 변화)**

  - 아이를 **덜 낳는 것이 아니라 늦게 낳는 것**

  - 실제코호트가 평생 낳는 자녀 수는 변하지 않을 수 있음

- **Quantum Effect (출산 수의 변화)**

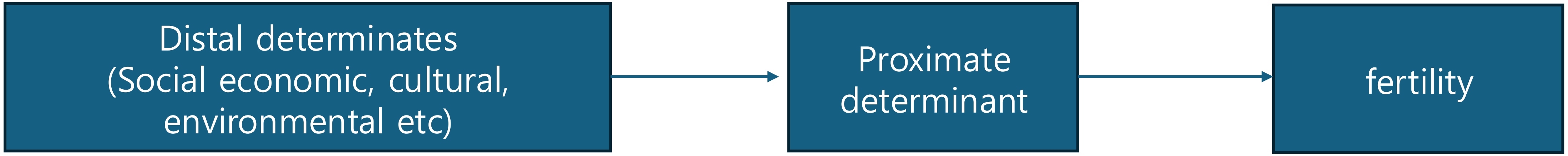
  - 실제로 **낳는 자녀 수가 감소**

  - 출산 곡선의 높이가 낮아짐

  - TFR 감소가 실제 출산 감소를 반영

Determinants of fertility

- 사회·경제적 변화는 어떻게 실제 출산으로 연결되는가?

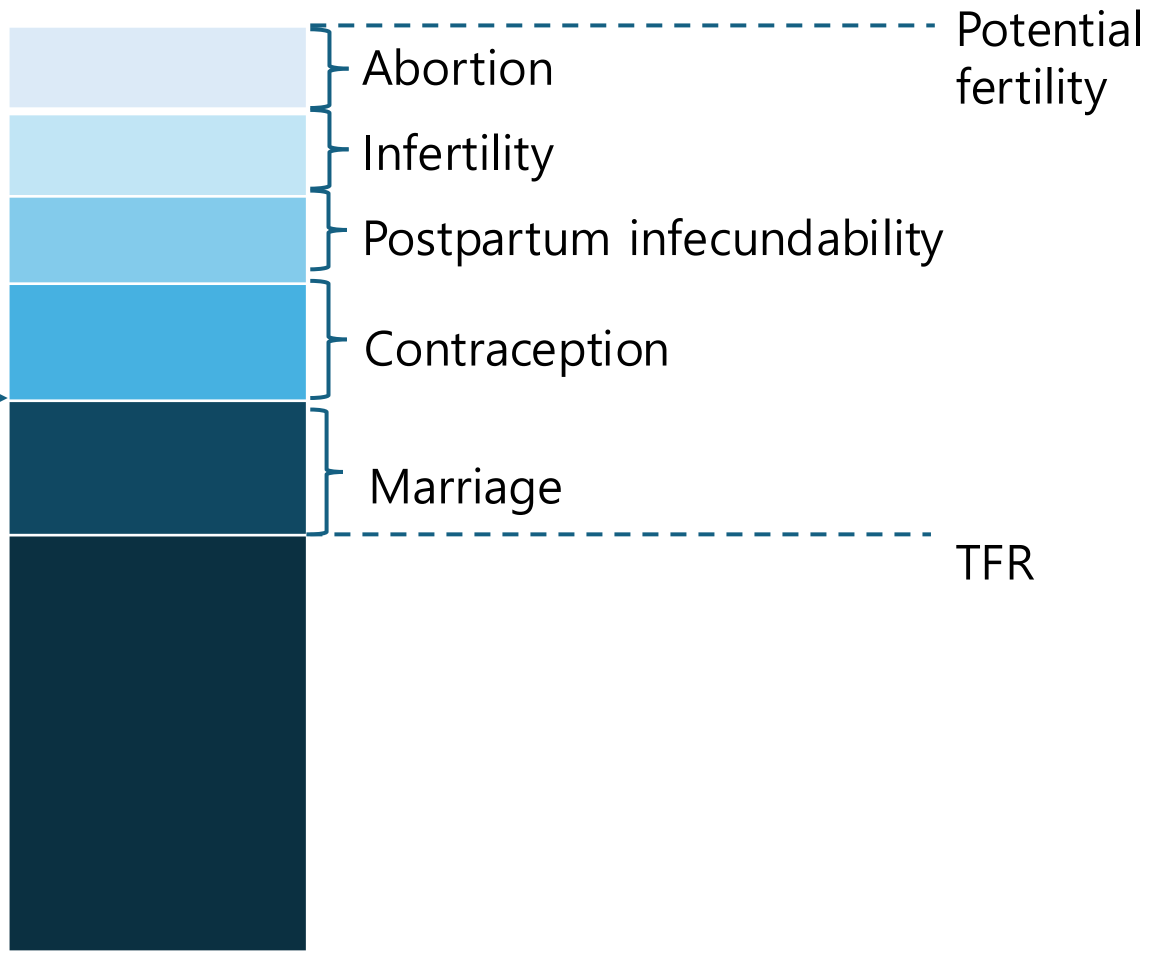


Bongaarts, J., & Potter, R. E. (2013). *Fertility, biology, and behavior: An analysis of the proximate determinants*. Academic press.

Bongaart's Proximate determinants

- Proximate Determinants 모두 출산을 감소시키는 방향(-)으로 작용
- Distal determinants(교육, 도시화, 여성취업, 문화 등)는 이 근접요인을 통해 출산에 영향을 미침

Distal determinates
(Social economic, cultural, environmental etc)



Q: 왜 어떤 나라는 TFR이 6명이고, 어떤 나라는 1명인가?

A: 여성 교육이 결혼·피임·출산연기 등을 변화시켰기 때문

Emerging Proximate determinants (MAR)

- **Medically Assisted Reproduction (MAR)**

  - Unlike traditional proximate determinants, MAR increases fertility rather than reducing it.

- **Why is MAR becoming important?**

  - 출산지연 글로벌화 (만혼 및 출산연기, 고령 출산 증가, 난임 유병률 증가)

  - Policy support (난임 시술 지원 확대, 인구·보건 빅데이터 활용 증가)

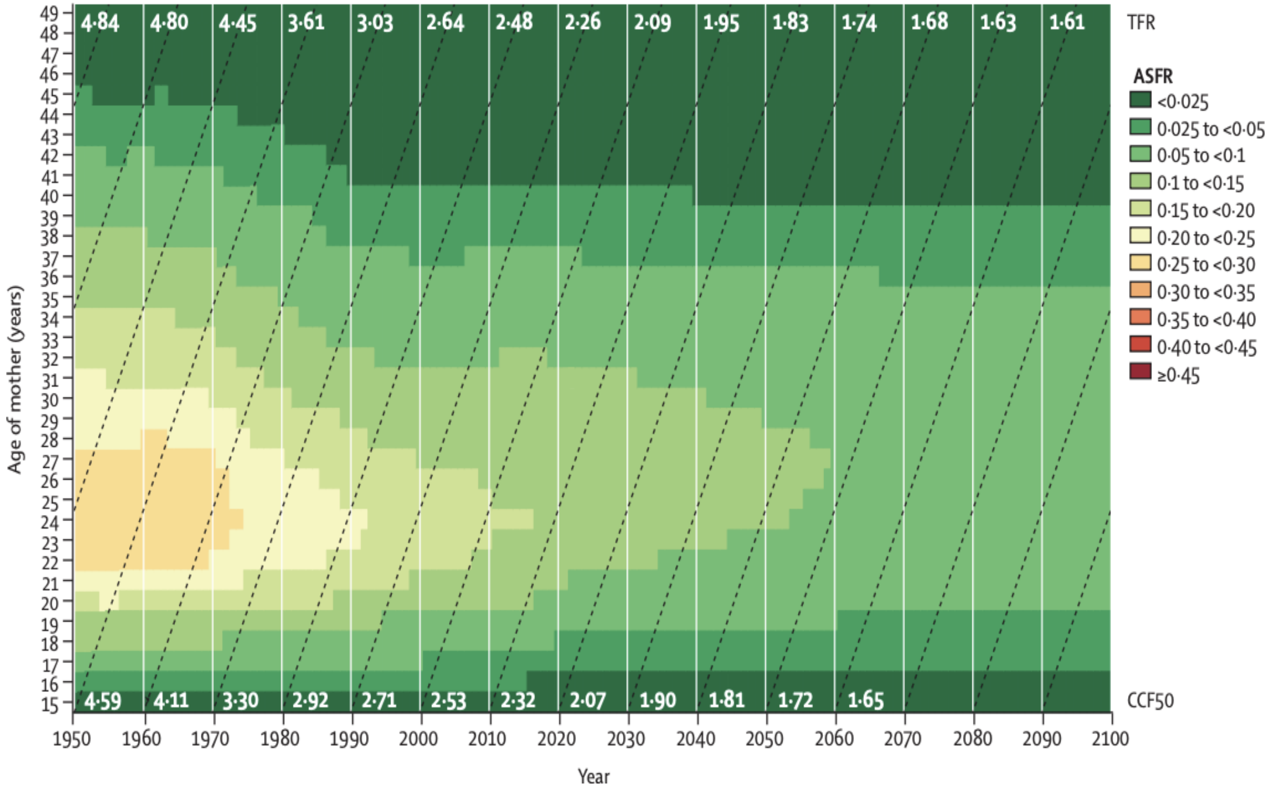
  - Demographic significance: 한국 전체 출생아의 약 10–12%가 MAR을 통해 출생

Age specific fertility in Global

	Age-specific fertility rate (livebirths per 1000 women annually)									Total fertility rate	Total fertility rate under age 25 years	Total fertility rate from ages 30 to 54 years	Number of livebirths	Net reproductive rate
	10–14 years	15–19 years	20–24 years	25–29 years	30–34 years	35–39 years	40–44 years	45–49 years	50–54 years					
Global	**0·81 (0·35–1·69)**	**42·9 (38·6–48·0)**	**129·4 (117·8–142·7)**	**131·9 (125·9–138·7)**	**96·8 (91·4–102·9)**	**52·4 (47·7–57·5)**	**17·2 (15·4–19·2)**	**3·4 (3·1–3·8)**	**0·06 (0·06–0·06)**	**2·4 (2·2–2·5)**	**0·87 (0·78–0·96)**	**0·85 (0·79–0·92)**	**138 810 622 (129 960 385–149 058 367)**	**1·08 (1·02–1·16)**
Low SDI	1·4 (0·6–2·9)	71·9 (65·0–80·0)	202·6 (181·7–225·9)	188·4 (177·3–200·9)	147·0 (136·9–158·2)	93·4 (83·6–103·5)	44·6 (39·5–49·9)	15·1 (13·7–16·7)	0·29 (0·28–0·3)	3·8 (3·6–4·1)	1·4 (1·2–1·5)	1·5 (1·4–1·6)	37 891 965 (35 159 071–41 108 482)	1·68 (1·58–1·81)
Low-middle SDI	0·88 (0·39–1·85)	51·7 (45·5–59·3)	156·6 (141·0–174·3)	152·0 (143·4–161·4)	112·4 (104·4–121·8)	63·8 (57·0–72·0)	24·1 (21·0–27·7)	6·4 (5·5–7·5)	0·12 (0·11–0·12)	2·8 (2·6–3·1)	1·0 (0·9–1·2)	1·0 (0·9–1·1)	40 394 490 (37 088 216–44 296 344)	1·28 (1·18–1·4)
Middle SDI	0·61 (0·27–1·27)	33·5 (30·1–37·6)	112·4 (100·6–125·9)	120·1 (113·3–127·9)	79·2 (73·7–85·3)	39·0 (34·4–44·2)	11·3 (9·9–13·1)	1·4 (1·1–1·6)	0·03 (0·03–0·03)	2·0 (1·8–2·2)	0·73 (0·66–0·82)	0·65 (0·6–0·72)	26 502 966 (24 536 281–28 871 941)	0·83 (0·77–0·9)
High-middle SDI	0·42 (0·18–0·86)	19·8 (18·3–21·6)	84·2 (79·0–89·9)	107·0 (103·4–110·7)	69·4 (65·8–73·1)	32·1 (29·2–35·3)	8·0 (7·2–8·8)	0·63 (0·53–0·74)	0·01 (0·01–0·01)	1·6 (1·5–1·7)	0·52 (0·49–0·56)	0·55 (0·51–0·59)	22 028 156 (20 983 021–23 184 413)	0·86 (0·81–0·9)
High SDI	0·25 (0·11–0·5)	12·5 (11·3–14·0)	49·6 (44·4–55·7)	89·5 (84·3–95·2)	98·6 (91·2–106·8)	51·8 (45·2–59·4)	11·1 (9·4–13·2)	0·63 (0·55–0·72)	0·01 (0·01–0·01)	1·6 (1·4–1·7)	0·31 (0·28–0·35)	0·81 (0·73–0·9)	11 638 396 (10 631 265–12 780 564)	0·76 (0·69–0·83)

Murray, Christopher JL, et al. "Population and fertility by age and sex for 195 countries and territories, 1950–2017: a systematic analysis for the Global Burden of Disease Study 2017." *The Lancet* 392.10159 (2018): 1995-2051.

Future Fertility



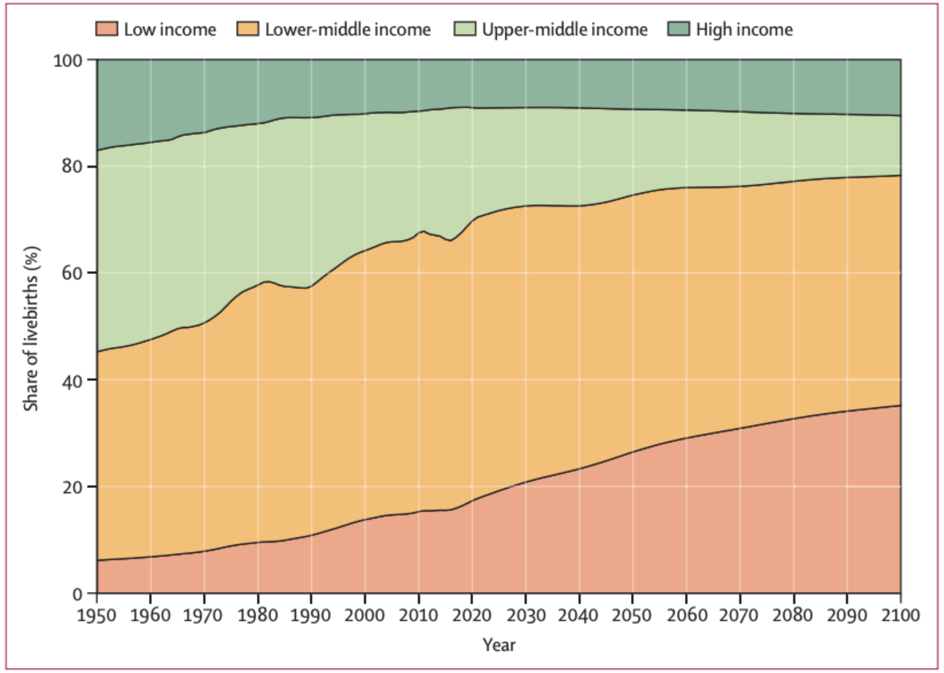


***Figure 5:* Share of livebirths by 2021 World Bank income group**
The World Bank income group is explicitly chosen to highlight the share of births according to resources per person. Resources are defined as gross national income (gross domestic product plus net income) per person.

Bhattacharjee, Natalia V., et al. "Global fertility in 204 countries and territories, 1950–2021, with forecasts to 2100: a comprehensive demographic analysis for the Global Burden of Disease Study 2021." *The lancet* 403.10440 (2024): 2057-2099.

Conclusion

1. **사망(Mortality)**
   - 사망률 감소 → 기대수명 증가
   - 생존(survival) 중심 지표에서 건강(health) 중심 지표로 발전
   - LMICs(YLL) + HIMc(YLD)
   - 감염병 감소 but pandemic
   - 기대수명증가 & NCD 증가 (중저소득국가의 Double Burden)

2. **출산(Fertility):**
   - 출산감소 & 출산지연
   - 저출산 만연: 많은 LMICs TFR 2.1 이하 & 주로 SSAs에서 TFR 2.1 이상 유지
   - 최근에는 MAR(난임시술)이 새로운 출산 결정요인으로 등장
   - 고령 출산 증가 & MAR 확대

감사합니다